

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned the correct position in the FIFO queue. Fail: A user who has reached the maximum allowed number of books can place another reservation.	Core functionality for reserving unavailable books.	
FR-002	Functional	The system shall allow Student Members to view the availability status and details of books in the library catalogue.	High	Pass: Searching for a valid ISBN displays the book title, author, ISBN, and current availability status. Fail: Any required book information is missing or incorrect.	Students need book information before deciding whether to reserve it.	
FR-003	Functional	The system shall automatically notify the next Student Member in the holding queue when a reserved book becomes available.	High	Pass: The next eligible member receives a notification when the book becomes available. Fail: The notification is not generated or is sent to the wrong queue member.	Ensures users know when their reserved book is available.	

FR-004	Functional	The system shall allow the Head Librarian to manage book holding queues and update the status of reserved books.	Medium	Pass: The Head Librarian can view, update, and remove queue entries and change a book's reservation status. Fail: Any authorized queue/status operation cannot be completed.	Librarians need administrative control over reservations and queues.	
FR-005	Functional	The system shall record and allow the Head Librarian to track overdue fine transactions associated with library members.	Medium	Pass: An overdue fine is recorded with the member, amount, and transaction status. Fail: The system fails to record or retrieve the fine transaction correctly.	Required to track overdue financial transactions.	
NFR-001	Nonfunctional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmark testing shows the query returns results in under 150 ms under the specified load. Fail: Response time is 150 ms or greater.	Ensures fast catalogue searching even with a large number of titles.	
NFR-002	Nonfunctional	The system shall protect library member and fine transaction data from unauthorized access.	High	Pass: Unauthorized users cannot access member or fine information during security testing. Fail: An unauthorized user can view or modify protected data.	Protects sensitive library and transaction information.	

